

Coordinate-Commensurable Subspaces of Spaces with Norming M-Bases

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Source Problem

Hajek and Russo ask whether admitting a norming Markushevich basis is hereditary to subspaces [1, Problem 6.1]. The problem appears on PDF page 24 of arXiv:2402.18442 and in the parsed source at `manuscript.tex:685-691`.

distinct conditions in general. In other words, to the best of the authors' knowledge, the following problem is still open.

Problem 6.1. *Let $\mathcal{X}$ be a Banach space with norming M-basis and $\mathcal{Z}$ be a subspace of $\mathcal{X}$. Must $\mathcal{Z}$ admit a norming M-basis as well?*

In this section we will discuss some particular cases where a positive answer is available

Result

This packet proves a general positive subcase: subspaces generated by subsets of a given norming M-basis inherit the corresponding restricted norming M-basis. Combining this with the earlier finite-codimensional invariance packet gives the finite-dimensional perturbation version below.

Theorem 1. *Let X be a Banach space with a λ -norming M-basis $\{e_\gamma; \varphi_\gamma\}_{\gamma \in \Gamma}$, where $0 < \lambda \leq 1$. For $A \subseteq \Gamma$, put*

$$X_A = \overline{\text{span}}\{e_\gamma : \gamma \in A\}.$$

Then

$$\{e_\gamma; \varphi_\gamma|_{X_A}\}_{\gamma \in A}$$

is a λ -norming M-basis of X_A .

Corollary 1. *Let X have a norming M-basis $\{e_\gamma; \varphi_\gamma\}_{\gamma \in \Gamma}$, and let Z be a closed subspace of X . Suppose that there is $A \subseteq \Gamma$ such that, for $X_A = \overline{\text{span}}\{e_\gamma : \gamma \in A\}$, the intersection*

$$V = Z \cap X_A$$

has finite codimension both in Z and in X_A . Then Z admits a norming M-basis.

Proof Intuition

The coordinate theorem is a useful tautness property of M-bases. If a vector is built from coordinates in A , then all coordinate functionals outside A vanish on it. Since the ambient coordinate span is already norming, its restriction to the coordinate subspace is exactly the smaller coordinate span indexed by A , and the norming inequality is preserved.

The corollary says that this survives finite-dimensional error. First apply the coordinate theorem to X_A . Then use the finite-codimensional heredity theorem from the earlier packet to pass from X_A to $V = Z \cap X_A$. Finally, since Z/V is finite-dimensional, append a finite biorthogonal system on a finite-dimensional complement.

Proof

Proof of Theorem 1. Biorthogonality is inherited from the ambient basis. The vectors $\{e_\gamma : \gamma \in A\}$ are fundamental in X_A by definition.

Let

$$D_A = \text{span}\{\varphi_\gamma|_{X_A} : \gamma \in A\} \subseteq X_A^*.$$

If $\eta \in \Gamma \setminus A$, then φ_η vanishes on $\text{span}\{e_\gamma : \gamma \in A\}$, hence by continuity

$$\varphi_\eta|_{X_A} = 0.$$

Therefore

$$\text{span}\{\varphi_\gamma : \gamma \in \Gamma\}|_{X_A} = D_A.$$

For $x \in X_A$, the ambient λ -norming inequality gives

$$\lambda\|x\|_X \leq \sup\{|y^*(x)| : y^* \in \text{span}\{\varphi_\gamma : \gamma \in \Gamma\}, \|y^*\|_{X^*} \leq 1\}.$$

Each restriction $y^*|_{X_A}$ belongs to D_A and has norm at most $\|y^*\|_{X^*}$. Hence the right hand side is bounded above by

$$\sup\{|g(x)| : g \in D_A, \|g\|_{X_A^*} \leq 1\}.$$

Thus D_A is λ -norming for X_A . In particular D_A is total, so the displayed biorthogonal fundamental system is a λ -norming M-basis of X_A . $\square$

Lemma 1 (Finite-codimensional inheritance). *If a Banach space Y has a norming M-basis and W is a closed finite-codimensional subspace of Y , then W has a norming M-basis.*

Proof. This is the main theorem of the earlier local finite-codimensional packet. The proof uses finite pivot elimination on coordinate annihilators and then reduces an arbitrary finite-codimensional subspace to that coordinate-annihilator case. $\square$

Proof of Corollary 1. By Theorem 1, X_A has a norming M-basis. Since $V = Z \cap X_A$ has finite codimension in X_A , Lemma 1 implies that V has a norming M-basis.

Because V has finite codimension in Z , choose a finite-dimensional complement L so that $Z = V \oplus L$, with bounded coordinate projections $P_V : Z \rightarrow V$ and $P_L : Z \rightarrow L$. Let $\{v_\alpha; v_\alpha^*\}_{\alpha \in I}$ be a λ -norming M-basis of V , and choose a finite biorthogonal basis $\{u_j; u_j^*\}_{j=1}^n$ for L .

Then

$$\{v_\alpha; v_\alpha^* \circ P_V\}_{\alpha \in I} \cup \{u_j; u_j^* \circ P_L\}_{j=1}^n$$

is a biorthogonal fundamental system in Z , and its functionals are total. Its span is norming because V^* is normed by $\text{span}\{v_\alpha^* : \alpha \in I\}$, L^* norms L , and the finite direct-sum norm $\|v\| + \|u\|$ is equivalent to the inherited norm on $Z = V \oplus L$. Therefore Z admits a norming M-basis. $\square$

Limitations and Novelty Check

This does not solve Problem 6.1 for arbitrary subspaces. It covers subspaces that are aligned with a chosen norming M-basis up to finite-dimensional error. Equivalently, a counterexample would have to be infinite-dimensionally skew relative to every ambient norming M-basis.

The coordinate theorem is elementary but appears useful as a bookkeeping principle for the heredity problem. It also explains the HRST coordinate-subspace packet as a special case. The commensurable corollary depends on the previously promoted finite-codimensional invariance theorem.

Human Review Recommendation

Review as a broad partial result and organizing lemma for Problem 6.1. The coordinate theorem is self-contained. For the commensurable corollary, check the finite-codimensional packet and the final finite-dimensional extension argument.

References

- [1] P. Hajek and T. Russo, *Norming Markushevich bases: recent results and open problems*, arXiv:2402.18442.